



AI In Business Decision Making: Transforming Organisational Intelligence In The Digital Era

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Abstract

Artificial Intelligence (AI) has emerged as a transformative force in contemporary business ecosystems, fundamentally reshaping how organisations collect, analyse, and act upon information to make strategic decisions. This paper investigates the multifaceted role of AI in business decision-making across operational, managerial, and strategic levels, with a focus on real-world adoption patterns between 2020 and 2025. Using a mixed-methods research design combining a quantitative survey of 280 business executives with qualitative case analyses of five leading organisations, the study reveals that AI-driven decision-making tools significantly enhance accuracy, speed, and cost-efficiency. Organisations integrating AI into their core decision frameworks report a 38.4% improvement in decision accuracy and a 31.7% reduction in decision-cycle time. However, key barriers, including algorithmic bias, data privacy concerns, workforce resistance, and infrastructure limitations, persist. The paper proposes a structured AI Decision Integration Framework (ADIF) as a roadmap for sustainable AI adoption. Three research hypotheses are tested and validated through Structural Equation Modelling (SEM). Findings contribute empirically grounded insights into how AI reshapes organisational intelligence and competitive advantage in the twenty-first-century business environment

Keywords: *Artificial Intelligence, Business Decision Making, Digital Transformation, Organisational Intelligence, Machine Learning, Strategic Management*



1. Introduction & Theoretical Framework

The twenty-first century has ushered in an era of unprecedented technological acceleration, with Artificial Intelligence (AI) standing at its epicentre. Three complementary theories underpin this study: Simon's (1955) Bounded Rationality Theory, the Technology Acceptance Model (Davis, 1989), and the Resource-Based View (Barney, 1991). The AI Decision Integration Framework (ADIF) comprises four pillars: Data Infrastructure, Algorithmic Governance, Workforce Enablement, and Leadership Alignment.



2. Research Methodology

A convergent parallel mixed-methods design was employed across five sectors-banking, retail, manufacturing, healthcare, and IT-in metropolitan and tier-1 Indian cities. The quantitative strand involved a structured questionnaire administered to 320 executives using stratified random sampling, yielding 280 usable responses (87.5% response rate), complemented by case study analyses of five organisations using NVivo 12.0, SPSS v26.0, and SmartPLS 3.0.



2. Research Methodology (Continued)

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3. Results & Discussion

All three hypotheses are supported, confirming that AI adoption positively and significantly influences both decision accuracy (H1: $\beta = 0.481$) and decision speed (H2: $\beta = 0.437$), while improved decision accuracy mediates the path to organisational performance (H3: $\beta = 0.392$). Companies that piloted all four ADIF pillars reported a 38.4% improvement in decision accuracy and 31.7% faster decision cycles.



4. Conclusions

This study advances understanding of AI's role in business decision-making by providing empirically grounded evidence from the Indian context. The three hypotheses were validated, and the proposed ADIF framework offers a practical roadmap for leveraging AI into sustainable competitive advantage.



4. Conclusions (Continued)

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